

Pen and Paper Encryption & Decryption

1. Encrypt "CHASE YOUR DREAMS"

a. Caesar Cipher

The standard Caesar cipher uses a forward shift of 3 (A becomes D, B becomes E, etc.).

Plaintext: C H A S E Y O U R D R E A M S

Shift +3: ↓↓↓↓↓ ↓↓↓ ↓↓↓↓↓

Ciphertext: F K D V H B R X U G U H D P V

Result: FKDVH BRXU GUHDPV

b. Shift Cipher with key 7

This uses a forward shift of 7 (A becomes H, B becomes I, etc.).

Plaintext: C H A S E Y O U R D R E A M S

Shift +7: ↓↓↓↓↓ ↓↓↓ ↓↓↓↓↓

Ciphertext: J O H Z L F V B Y K Y L H T Z

Result: JOHZL FVBY KYLHTZ

c. ROT 13

ROT13 replaces a letter with the 13th letter after it in the alphabet (A becomes N, B becomes O, etc.).

Plaintext: C H A S E Y O U R D R E A M S

Shift +13: ↓↓↓↓↓ ↓↓↓ ↓↓↓↓↓

Ciphertext: P U N F R L B H U Q E R N Z F

Result: PUNFR LBHU QERNZF

2. Decrypt

a. "WRWDOOB VHFUHW PHVVDJH" using Caesar cipher

To decrypt a standard Caesar cipher, we reverse the process with a backward shift of 3 (D becomes A, E becomes B, etc.).

Ciphertext: W R W D O O B V H F U H W P H V V D J H

Shift -3: ↓↓↓↓↓↓ ↓↓↓↓↓ ↓↓↓↓↓↓

Plaintext: T O T A L L Y S E C R E T M E S S A G E

Result: TOTALLY SECRET MESSAGE

b. "SJI IBTW YNRJ" using Shift Cipher, 5

To decrypt, we reverse the +5 encryption by applying a backward shift of 5 (F becomes A, G becomes B, etc.).

Ciphertext: S J I I B T W Y N R J

Shift -5: ↓↓↓ ↓↓↓ ↓↓↓

Plaintext: N E D D W O R T I M E

Result: NED DWOR TIME

(Note: If the intended plaintext was "NEED MORE TIME", the correct ciphertext with a shift of 5 would have been "SJJI RTWJ YNRJ".)

c. "FRAQ ZBER PBSSRR" using ROT13

Because the alphabet has 26 letters, shifting forward by 13 twice brings you back to the start. To decrypt ROT13, you simply apply ROT13 (+13 shift) again.

Ciphertext: F R A Q Z B E R P B S S R R

Shift +13: ↓↓↓↓ ↓↓↓↓ ↓↓↓↓↓↓

Plaintext: S E N D M O R E C O F F E E

Result: SEND MORE COFFEE